

CS 59300

Application of Deep Learning Introduction to Course

Fall 2025

Instructor: Dr. Zesheng Chen

Introduce Yourself

- What is your name (both last name and first name)?
- What is your expectation on this course?
- Seating chart

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A Quick Introduction to the Course

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Class information

Lecture:

- Instructor: Dr. Zesheng Chen
- Office hours: MTWR 1:30 - 2:45 PM or R 11:00 AM - noon
- Email: chen439@pfw.edu
- Ensure that you have access to the course web site at **Brightspace**
 - Announcements, Syllabus, Slides and Examples, Homework, Exam, and Grades
 - Email the instructor if you do not have access
- Grader: Prajwal Hosahalli Dayananda <hosap02@pfw.edu>

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Survey

<https://pollev.com/zeshengchen845>

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What is Deep Learning?

- Two different meanings
 - Learning methodology
 - Learn a subject in deep
 - Next lecture
 - One popular machine learning method
- Apply **deep learning** methodology in learning “the application of **deep learning**”

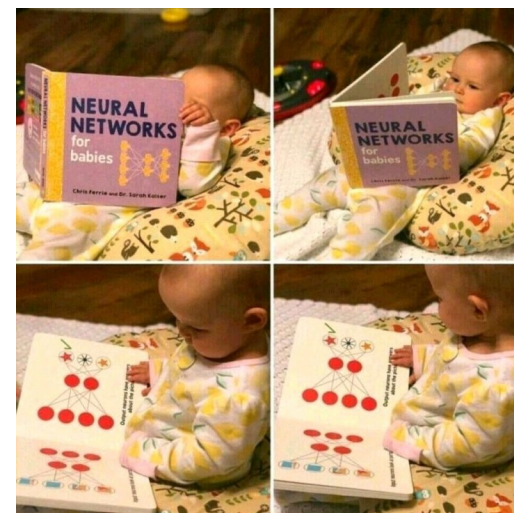
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Deep Learning (A Machine Learning Model)

- Main contents
 - Applications
 - ChatGPT and its APIs
 - Hugging Face
 - Models
 - ANN / CNN / RNN / Transformer / VAE / GAN
 - Research (optional)
 - Explainability
 - Security
 - Reinforcement learning

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Why Deep Learning?



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Why Deep Learning?

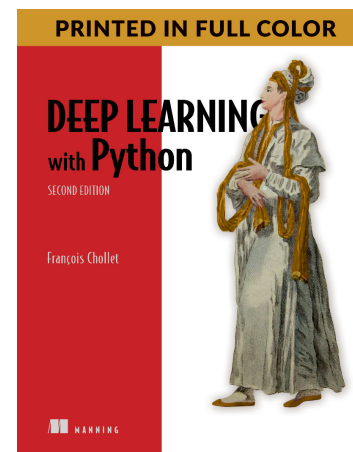


- https://www.youtube.com/watch?v=WXuK6gekU1Y&t=3s&ab_channel=DeepMind

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Reference Books

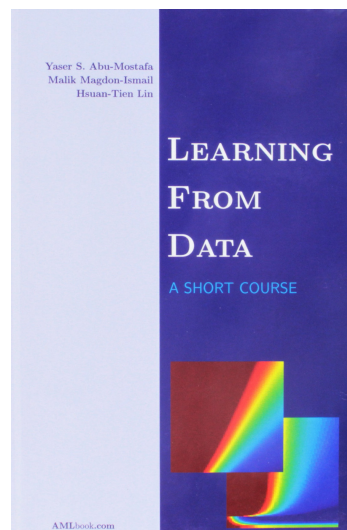
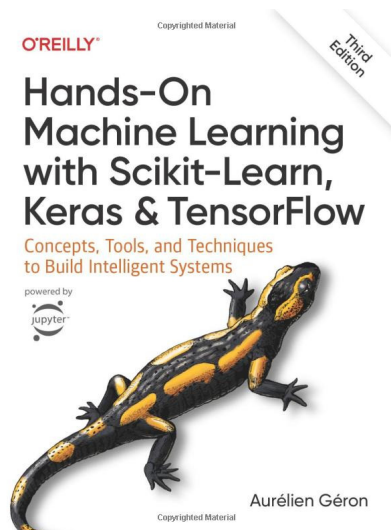


- <https://www.amazon.com/Learning-Python-Second-Fran%C3%A7ois-Chollet/dp/1617296864>

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Reference Books



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Grading

- Homework: 30%
- Course project: 30%
- Final exam: 30%
- Class participation: 10%

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Grading (more)

- A 90 – 100
- B 80 – 89
- C 70 – 79
- D 60 – 69
- F Below 60

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Homework

- Tentatively about 8 homework assignments
 - ☐ Usually one or two weeks to work on
 - ☐ Please start early!
- Homework 1 and 2
 - ☐ Take Kaggle courses
 - ☐ Learn or review Python and machine learning
 - ☐ Get certificates
- Some assignments are based on class examples

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Final Exam

- Open books, open notes
- Allow to use laptop
- Allow to access Google Colab
- **Not** allow to discuss with each other
- **Not** allow to use Google or other search engines or AI tools (e.g., ChatGPT)
- Tentatively on December 19, Friday, between 1pm and 3pm in the classroom

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Course Project

- A deep-learning based project your group choose (30%)
 - ☐ About **four** students in a group
 - ☐ Requirements
 - ☐ Planning
 - ☐ Reviews
 - ☐ Final presentation
 - ☐ Publish the code at your **GitHub**
 - “**GitHub is your resume!**”

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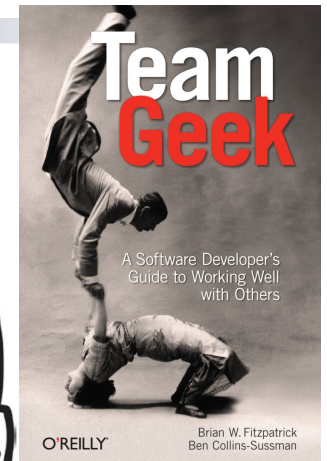


Course Project (Cont.)

- Course project milestones (tentatively):
 - (Sept. 3) Form the team (email the instructor the agreement sheet)
 - (Sept. 17) Project topic *
 - (Sept. 24) Requirements *
 - (Oct. 29) Project review *
 - (Dec. 8 / 10) Final presentation *
- * : Need to give a presentation in class

More information next lecture!

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Class Policies

- Homework must be done individually
- No homework will be accepted after the due date
- Academy Integrity Code in PFW student manual
 - **Cheating**
 - **Plagiarism**
- Missed exam regulations
- Class participation
- Late arrival

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Golden Rule

.....

“Do unto others as you would have them do unto you”

Matt. 7:12

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Acknowledgements

- Thank Dr. Hung-Yi Lee for allowing me to use his class materials
 - <https://speech.ee.ntu.edu.tw/~hylee/ml/2022-spring.php> (Machine Learning (Spring 2022))
 - <https://speech.ee.ntu.edu.tw/~hylee/genai/2024-spring.php> (Introduction to Generative AI (Spring 2024))
- Refer to Dr. Jeff Heaton's course materials
 - https://github.com/jeffheaton/app_generative_ai
 - https://github.com/jeffheaton/app_deep_learning

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0.1 Extra Point Policy

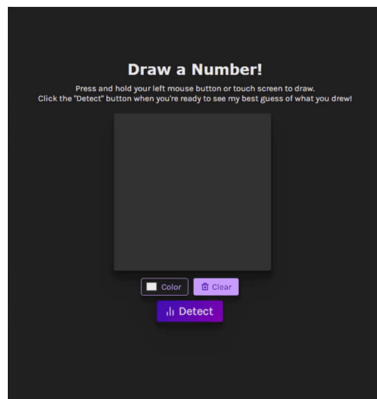


Up to 3 points towards the final grade!

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Deep Learning: “Hello World” Example



- <https://digit-inator.netlify.app/>
- <https://github.com/RodneyMcQuain/Digit-Inator>

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Q & A

Have fun!!!